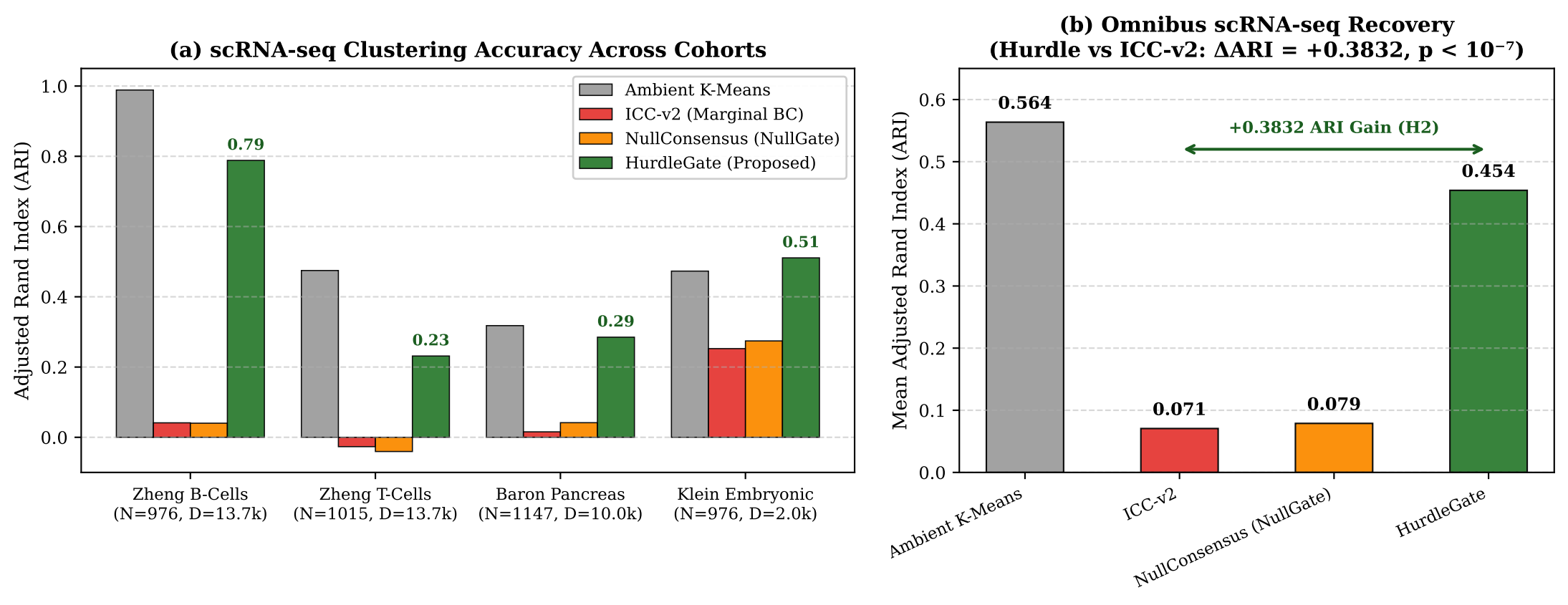
(a) scRNA-seq Clustering Accuracy Across Cohorts
Adjusted Rand Index (ARI)
1.0
0.8
0.6
0.4
0.2
0.0
Ambient K-Means
ICC-v2 (Marginal BC)
NullConsensus (NullGate)
HurdleGate (Proposed)
0.79
0.23
0.29
0.51
Zheng B-Cells
(N=976, D=13.7k)
Zheng T-Cells
(N=1015, D=13.7k)
Baron Pancreas
(N=1147, D=10.0k)
Klein Embryonic
(N=976, D=2.0k)
(b) Omnibus scRNA-seq Recovery
(Hurdle vs ICC-v2: ΔARI = +0.3832, p < 10⁻⁷)
Mean Adjusted Rand Index (ARI)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.564
+0.3832 ARI Gain (H2)
0.454
0.071
0.079
Ambient K-Means
ICC-v2
NullConsensus (NullGate)
HurdleGate